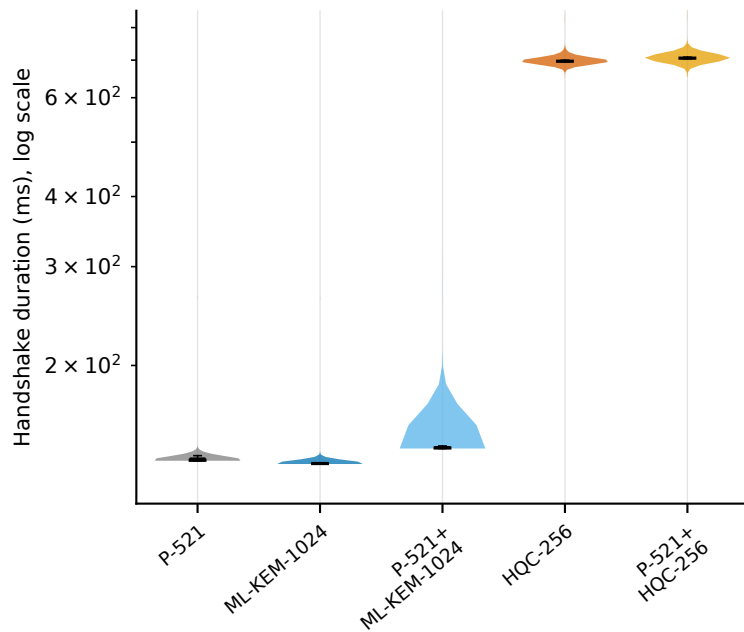
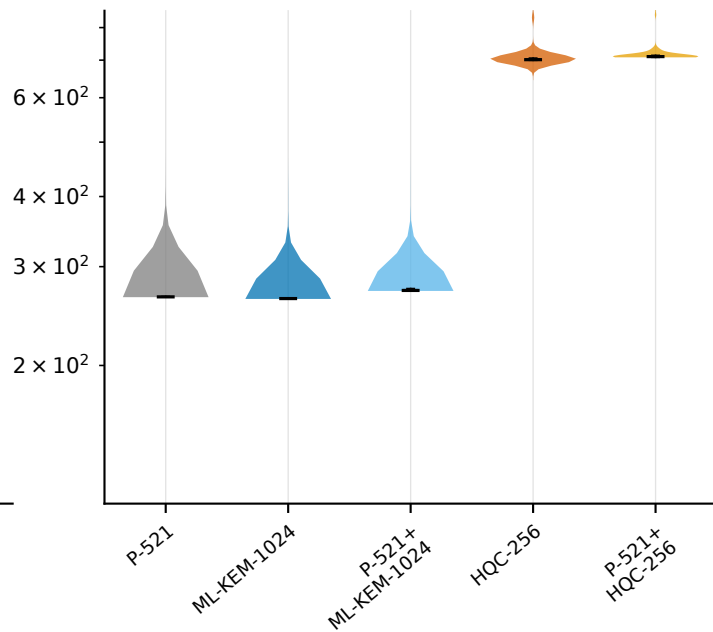


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	136.4	4.3	266.3	271.4	49.5	3.27k
ML-KEM-1024	134.1	4.4	264.6	268.0	37.8	2.52k
P-521+ML-KEM-1024	147.0	43.3	1.56k	277.8	36.4	2.53k
HQC-256	701.1	4.8	1.36k	709.6	6.6	1.63k
P-521+HQC-256	709.5	6.4	1.70k	716.4	3.4	938.0